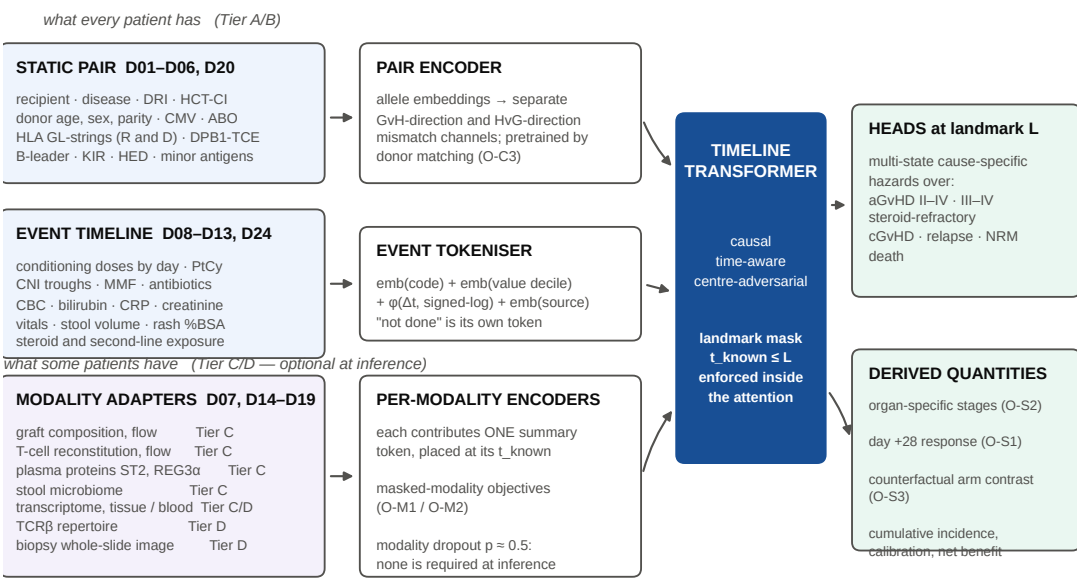
what every patient has (Tier A/B)
STATIC PAIR D01–D06, D20
recipient · disease · DRI · HCT-CI
donor age, sex, parity · CMV · ABO
HLA GL-strings (R and D) · DPB1-TCE
B-leader · KIR · HED · minor antigens
EVENT TIMELINE D08–D13, D24
conditioning doses by day · PtCy
CNI troughs · MMF · antibiotics
CBC · bilirubin · CRP · creatinine
vitals · stool volume · rash %BSA
steroid and second-line exposure
what some patients have (Tier C/D — optional at inference)
MODALITY ADAPTERS D07, D14–D19
graft composition, flow Tier C
T-cell reconstitution, flow Tier C
plasma proteins ST2, REG3α Tier C
stool microbiome Tier C
transcriptome, tissue / blood Tier C/D
TCRβ repertoire Tier D
biopsy whole-slide image Tier D
PAIR ENCODER
allele embeddings → separate GvH-direction and HvG-direction mismatch channels; pretrained by donor matching (O-C3)
EVENT TOKENISER
emb(code) + emb(value decile) + φ(Δt, signed-log) + emb(source)
"not done" is its own token
PER-MODALITY ENCODERS
each contributes ONE summary token, placed at its t_known
masked-modality objectives (O-M1 / O-M2)
modality dropout p ≈ 0.5: none is required at inference
TIMELINE TRANSFORMER
causal
time-aware
centre-adversarial
landmark mask t_known ≤ L enforced inside the attention
HEADS at landmark L
multi-state cause-specific hazards over:
aGvHD II–IV · III–IV
steroid-refractory
cGvHD · relapse · NRM
death
DERIVED QUANTITIES
organ-specific stages (O-S2)
day +28 response (O-S1)
counterfactual arm contrast (O-S3)
cumulative incidence, calibration, net benefit